Donggeon (David) Oh

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Research Interests

I push the boundaries of **safety assurance** for intelligent systems. In doing so,

- I integrate insights from **control theory**, **reinforcement learning (RL)**, and **dynamic game theory** to certify and enforce safety of high-dimensional, black-box systems under uncertainty.
- I investigate the **interplay between humans and robots** in safety-critical scenarios and how to foster safe, smooth, and **trustworthy collaboration**.

Ultimately, I envision a world where robots operate safely amidst other agents, including humans, in **unstructured, open-world environments**.

Education

Princeton University 2024–present
Ph.D. candidate in Electrical and Computer Engineering; expected May 2029.
Advisor: Jaime Fernández Fisac.

Seoul National University 2017–2024
B.S. in Aerospace Engineering & B.S. in Artificial Intelligence. Ranked 2nd of 71 students.

Publications

* Equal contribution. † Equal advising. ‡ Equal senior authorship.

Preprints

[P2] Winning a won game: Strict reach–avoid–stay control barrier functions for high-dimensional black-box systems
D. D. Oh, D. P. Nguyen, G. Yuan, Q. Li, J. F. Fisac†, H. Hu†
arXiv:2609.19449, 2026

[P1] Turning safety into competence: Minimally exploitable robot policies via safety-filtered reinforcement learning
R. Wu*, R. Yang*, **D. D. Oh**, D. P. Nguyen†, H. Hu†
arXiv:2609.27312, 2026

Peer-Reviewed Conference Publications

[C7] Synthesis and deployment of maximal robust control barrier functions through adversarial reinforcement learning
D. D. Oh, D. P. Nguyen, H. Hu, J. F. Fisac
*IEEE Conference on Decision and Control (**CDC**)*, 2026 *(to appear)*

[C6] Unified Overview of Safety Certificate Functions: Techniques from Hamilton–Jacobi Reachability and Control Barrier Functions
H. Wang, M. Cohen, Z. Feng, **D. D. Oh***, O. So*, S. Zhang*, J. Borquez, C. Fan†, J. F. Fisac†, H. Hu†, J. J. Choi‡, S. Bansal‡
*IEEE Conference on Decision and Control (**CDC**)*, 2026 *(Tutorial paper; to appear)*

[C5] Provably optimal reinforcement learning under safety filtering
D. D. Oh*, D. P. Nguyen*, H. Hu, J. F. Fisac
*Proceedings of **IASEAI** Conference*, 2026

[C4] Safety with agency: Human-centered safety filter with application to AI-assisted motorsports
D. D. Oh*, J. Lidard*, H. Hu, H. Sinhmar, E. Lazarski, D. Gopinath, E. S. Sumner, J. A. DeCastro, G. Rosman, N. E. Leonard, J. F. Fisac
*Robotics: Science and Systems (**RSS**)*, 2025

[C3] Safety-critical control under multiple state and input constraints and application to fixed-wing UAV
D. D. Oh*, D. Lee*, H. J. Kim
*IEEE Conference on Decision and Control (**CDC**)*, 2023

[**C2**] Stable contact guaranteeing motion/force control for an aerial manipulator on an arbitrarily tilted surface
J. Byun, B. Kim, C. Kim, **D. D. Oh**, H. J. Kim
*IEEE International Conference on Robotics and Automation (**ICRA**)*, 2023

[**C1**] Real-time trajectory generation of a quadrotor UAV with load suspended from a pulley
D. D. Oh, J. Byun, D. Lee
*International Conference on Control, Automation and Systems (**ICCAS**)*, 2022

Professional Activities

Workshop & Seminar Organization

[**W1**] Workshop on safety analysis for autonomous system: Practical perspectives and recent advances
H. Wang, S. Zhang, O. So, J. F. Fisac, C. Fan, S. Bansal, Z. Feng, **D. D. Oh**, J. J. Choi, M. Cohen, H. Hu, J. Borquez
*IEEE Conference on Decision and Control (**CDC**)*, 2026 *(upcoming)*

Reviewer

IEEE Robotics and Automation Letters (RA-L); IEEE Conference on Decision and Control (CDC); International Association for Safe & Ethical AI (IASEAI); International Conference on Neuro-symbolic Systems (NeuS); ICRA Workshop on Public Trust in Autonomous Systems (PTAS).

Invited Talks

[**T1**] Towards safety for black-box systems: Application to AI-assisted racing and safe RL
Safe and Intelligent Autonomy (SIA) Lab, **Stanford University**, Sept. 8, 2026.

[**T2**] Safety for black-box systems: Q-CBF theory and future directions
Alliance AI Lab, **Johns Hopkins University**, Oct. 21, 2026.

Patent Applications

[**PA1**] Safety system for a vehicle
N. Leonard, J. DeCastro, G. Rosman, J. F. Fisac, **D. D. Oh**, J. Lidard, H. Hu, D. Nguyen
U.S. Patent Application No. 19/641,805. Filed Apr. 2026. *Patent pending.*

Teaching & Mentoring

Head Teaching Assistant, Intelligent Robotic Systems — Spring 2026
Princeton University. Instructor: Jaime Fernández Fisac.

Research mentorship

Elle Lazarski (mentored as a Princeton undergraduate; now a Ph.D. student at UC Berkeley).

Honors & Awards

Presidential Science Scholarship, Republic of Korea — 2017–2022
National scholarship recognizing 120 top STEM students.

Talent Award of Korea — 2016
National honor recognizing 100 outstanding Koreans across disciplines.

Research Methods & Tools

Computation: Python, PyTorch, MATLAB/Simulink, C/C++, Linux, Git; MuJoCo, Safety Gymnasium.
Human-in-the-loop research: Authored IRB protocol and study materials and conducted an 83-participant driving study in Assetto Corsa; wheel/pedal interface with a custom Linux force-feedback torque driver.

References

Jaime Fernández Fisac, Princeton University — jfisac@princeton.edu
Assistant Professor of Electrical and Computer Engineering.

Naomi Ehrich Leonard, Princeton University — naomi@princeton.edu
Chair and Edwin S. Wilsey Professor of Mechanical and Aerospace Engineering.

Haimin Hu, Johns Hopkins University haimin@cs.jhu.edu
Assistant Professor of Computer Science.